

Christopher Bülte

DEEP LEARNING · PROBABILISTIC MODELING · UNCERTAINTY QUANTIFICATION
Ludwig-Maximilians University Munich, Akademiestraße 7, Munich, 80799 Germany
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Objective

Aspiring researcher in theoretical and physics-informed machine learning with a strong background in statistics and data science, always eager to learn something new. Special interest in foundations of deep learning, uncertainty quantification, statistics, and machine learning for the sciences.

Education

Ludwigs-Maximilians-University Munich (LMU)

Munich, Germany

PHD ON MATHEMATICAL FOUNDATIONS OF DEEP LEARNING

03/2024 - present

- Research on the intersection of machine learning, uncertainty quantification and probabilistic modeling
- Applying novel probabilistic machine learning methods to the engineering and natural sciences
- Associated PhD student, Konrad Zuse School of Excellence in Reliable AI and Munich Center for Machine Learning
- Coordinator of the DFG priority program: Theoretical foundations of deep learning (SPP2298)
- Advisor: Prof. Dr. Gitta Kutyniok

Karlsruhe Institute of Technology (KIT)

Karlsruhe, Germany

M.SC. IN ECONOMATHEMATICS

04/2021 - 02/2024

- Grade: 1.1/5 (1 highest, 5 lowest, Top 2%)
- Thesis: *Estimation of Extremes for Spatio-Temporal Processes with Neural Networks*, Advisor: Prof. Dr. Melanie Schienle
- Supplementary studies on Sustainable Development

Karlsruhe Institute of Technology (KIT)

Karlsruhe, Germany

B.SC. IN INDUSTRIAL ENGINEERING

10/2017 - 03/2021

- Grade: 1.5/5 (1 highest, 5 lowest, Top 10%)
- Thesis: *Nonlinear Kernel Regression: Theoretical Aspects and Robust Extensions*

Professional experience

Chair of Statistical Methods and Econometrics, KIT

Karlsruhe, Germany

RESEARCH ASSISTANT

06/2022 - 02/2024

- Developed and analyzed neural network-based methods for large-scale probabilistic weather forecasting
- Implemented a webscraper for tracking health statistics for infectious disease modeling

anacision GmbH

Karlsruhe, Germany

DATA SCIENCE INTERN

09/2019 - 02/2020

- Forecasting and extreme value identification for large-scale time series (Python)
- Developed, tested and implemented algorithms for big data integration (Python, Apache Spark)

Karlsruhe Institute of Technology

Karlsruhe, Germany

ASSISTANT TEACHER

09/2018 - 02/2020

- Assistant Teacher for the courses Mathematics 1 & 2, and Statistics
- Conducted tutorials and problem-solving sessions for undergraduate students

Publications

JOURNAL & CONFERENCE PUBLICATIONS

Bülte, C., Scholl, P., and Kutyniok, G. Probabilistic neural operators for functional uncertainty quantification. **Transactions on Machine Learning Research**, 2025, [link]

Bülte, C., Horat, N., Quinting, J. and Lerch, S. Uncertainty quantification for data-driven weather models. **Artificial Intelligence for the Earth Systems**, 2025, doi: 10.1175/AIES-D-24-0049.1

Bülte, C., Kleinebrahm, M., Yilmaz, Ü. and Gomez-Rómero, J. Multivariate time series imputation for energy data using neural networks. **Energy and AI**, 2023, Vol. 13, doi: 10.1016/j.egyai.2023.100239

PEER-REVIEWED WORKSHOP PAPERS

Sale, Y., **Bülte, C.**, Czaja, F., Stiller, J., and Hüllermeier, E. What Uncertainties Do We Need for Dynamical Systems?. ICML 2026 Workshop on Epistemic Intelligence in Machine Learning.

Bülte, C., Sale, Y., Kutyniok, G., and Hüllermeier, E. A Formal Assessment of Uncertainty Measures in Regression. EurlPS 2025 Workshop on Epistemic Intelligence in Machine Learning (oral).

Kneissl, C., **Bülte, C.**, Scholl, P., Kutyniok, G. Uncertainty-aware diffusion models for probabilistic regression. EurlPS 2025 Workshop on Epistemic Intelligence in Machine Learning (oral).

Bülte, C., Maskey, S., Scholl, P., von Berg, J. and Kutyniok, G. Graph Neural Networks for Enhancing Ensemble Forecasts of Extreme Rainfall. ICLR 2025 Workshop on Tackling Climate Change with Machine Learning, [link]

Bülte, C., Scholl, P., and Kutyniok, G. Probabilistic predictions with Fourier neural operators. NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty, [link]

Yilmaz, Ü., Kleinebrahm, M., **Bülte, C.** and Gomez-Rómero, J. Applying transformer to imputation of multivariate energy time series data. ICML 2021 Workshop on Tackling Climate Change with Machine Learning, [link]

PREPRINTS

Bülte, C., Sale, Y., Kutyniok, G., and Hüllermeier, E. Uncertainty Quantification for Regression: A Unified Framework based on kernel scores. preprint, arxiv: 2510.25599

Kneissl, C., **Bülte, C.**, Scholl, P., Kutyniok, G. Improved probabilistic regression using diffusion models. preprint, arxiv: 2510.04583

Bülte, C., Sale, Y., Löhr, T., Hofman, P., Kutyniok, G., and Hüllermeier, E. An Axiomatic Assessment of Entropy- and Variance-based Uncertainty Quantification in Regression. preprint, arxiv: 2504.18433

Bülte, C., Leimenstoll, L., and Schienle, M. Modeling Spatial Extremal Dependence of Precipitation Using Distributional Neural Networks. preprint, arxiv: 2407.08668

Scientific Events

Workshop on Epistemic Intelligence in Machine Learning, Kopenhagen (Denmark), Oral presentation on *A Formal Assessment of Uncertainty Measures in Regression*.

MIAPbP Workshop: Build Big or Build Smart: Examining Scale and Domain Knowledge in Machine Learning for Fundamental Physics, Munich (Germany), Contributed talk on *Probabilistic neural operators*.

ICLR 2025 Workshop on Tackling Climate Change with Machine Learning, Singapore, Poster presentation on *Graph Neural Networks for Enhancing Ensemble Forecasts of Extreme Rainfall*.

NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty, Vancouver (Canada), Poster presentation on *Probabilistic predictions with Fourier neural operators*.

Compstat 2024, Gießen (Germany), Contributed talk on *Estimation of spatio-temporal extremes via generative neural networks*.

MathSEE Symposium 2023, Karlsruhe (Germany). Poster presentation on *Probabilistic data-driven weather forecasting* (Best poster award).

Community Services

REVIEWING & ORGANIZATION

2025 - 2026 Reviewer for Artificial Intelligence for the Earth Systems (AIES)

2025 - 2026 Organizer of DFG-Minisympodium at the Annual Meeting of GAMM

2025 - 2026 Program Committee at Epistemic Intelligence in Machine Learning Workshop (EIML)

2025 Reviewer for International Conference on Learning Representations (ICLR)

Other

Honors	Associate PhD relAI & MCML, Deutschlandstipendium 2019
Languages	German (native), English (fluent)
Coding	Python, git, LaTeX, R, Linux, Bash, High-performance computing
Coding Frameworks	PyTorch, TensorFlow, SciPy, Xarray, Weights & Biases
Past Engagements	EduRef e.V. Education for Refugees